

Part 1- Overall Information

Proposal Form for Advantech AIoT InnoWorks			
University Name	Politeknik Perkapalan Negeri Surabaya (PPNS)	Department/Institute	
Professor Name (Applicant/Host)	Team 1 : Zindhu Maulana Ahmad Putra Team 2 : Ryan Yudha Adhitya, S.ST., M.T. Team 3 : Dimas Pristovani Rinanda S.ST.,M.T		
Email			
Overall Information and Resource Requirement for AIoT InnoWorks Program			
Duration of AIoT InnoWorks Please specify the timeline of skills developments, application development and final demo. The suggested overall duration is 4-6 months.	● Skills developments: 2026/05/05 to 2026/05/16 ● Application development: 2026/MM/DD to YYYY/MM/DD ● Final Demo Day: YYYY/MM/DD		
How many number of students participate? Please include: 1. Number of teams 2. Number of team members in each teams	Team 1: [Pak Zindhu] 1. Muhammad Athaya Akhdan 2. Muchammad Rasya Rizky Pratama 3. Ahmad Zuhri Fahrizal		
How many slots on WISE-PaaS's training websites are needed for participants?	There are 2 training slots required by participants		
The duration of WISE-PaaS space open for application developments and final demo? For example: 2021 Apr 1~2021 Jun 30	YYYY/MM/DD		

Part 2- Project Information

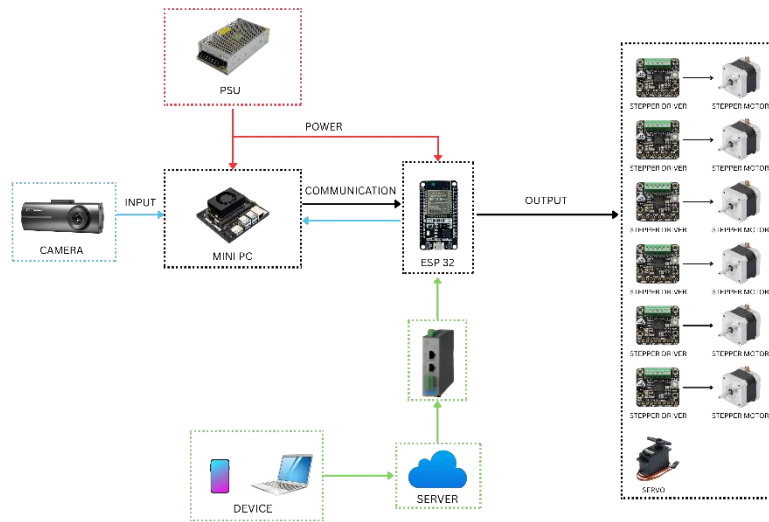
Please fill in project information for each team. The form can be copied for use.

Advantech AIoT InnoWorks - Project Information: Team 1			
Team Name		No. of Team Members	1
Project Topic	Development of a Software-Driven Object Identification and Transfer System Using YOLO and Robotic Arm Control	Adviser	Zindhu Maulana Ahmad Putra
Member background (department)	Automation Engineering / Ship Electrical Engineering		

Project introduction For example: The target audience of the project, the issue want to be resolved and the expected benefits of the project	This project proposes an AI-based object sorting and handling system using YOLO object detection and a robotic arm as the execution media. The system is designed to support automated item identification, classification, and placement through a software-driven workflow. The main target users of this project are educational laboratories, automation training centers, small-scale warehouses, and smart manufacturing learning environments that require a practical demonstration of computer vision and robotic automation. This project is also relevant to the current development of intelligent automation, where software is expected not only to monitor a process, but also to make decisions and control physical devices accurately. In many manual sorting and handling activities, operators still need to identify objects visually and move them to the appropriate location by hand. This process is time-consuming, inconsistent, and vulnerable to human error, especially when different types of items must be processed repeatedly. In addition, conventional manual handling systems usually do not provide real-time monitoring, detection history, or automatic decision-making support. These limitations show the need for a more adaptive system that can recognize objects automatically, reduce dependency on manual operation, and provide traceable operational data. To address these issues, this project develops a software system that integrates camera-based image acquisition, YOLO object detection, object classification, decision logic, robotic arm command generation, and dashboard monitoring. The camera captures the object placed in the input area, while the YOLO model detects and classifies the object type. After the object is recognized, the software determines the correct destination and sends the movement command to the robotic arm controller. The robotic arm then picks up the object and places it in the assigned target area. The use of YOLO is selected because it supports real-time object detection, making it suitable for applications that require fast recognition and direct response from the robotic system. The strength of this project lies in its software architecture. The system does not only move objects mechanically, but also connects visual perception, classification results, decision-making logic, and robotic execution into one integrated workflow. This makes the project scalable, because the object categories, placement rules, dashboard features, and detection model can be updated or expanded without redesigning the entire hardware system. Therefore, the proposed system can be further developed for various use cases, such as educational robotics, automated sorting, warehouse assistance, quality inspection, and smart manufacturing simulation. The expected benefit of this project is to improve the efficiency, accuracy, and consistency of object sorting and handling activities. The system also provides real-time information such as object category, detection result, robotic arm status, and operation history through a monitoring dashboard. By focusing on the integration between artificial intelligence, software control, and robotic execution, this project offers a practical solution for learning and implementing intelligent automation in modern industrial and educational environments. Overall, this project demonstrates how software development can become the core intelligence of an automation system, while the robotic arm functions as a physical medium to execute software-based decisions.
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Please provide the functional/system diagram of the project and briefly describe how WISE-PaaS will be utilized in the project?

For example: Which functional modules of WISE-PaaS are expected to be used?

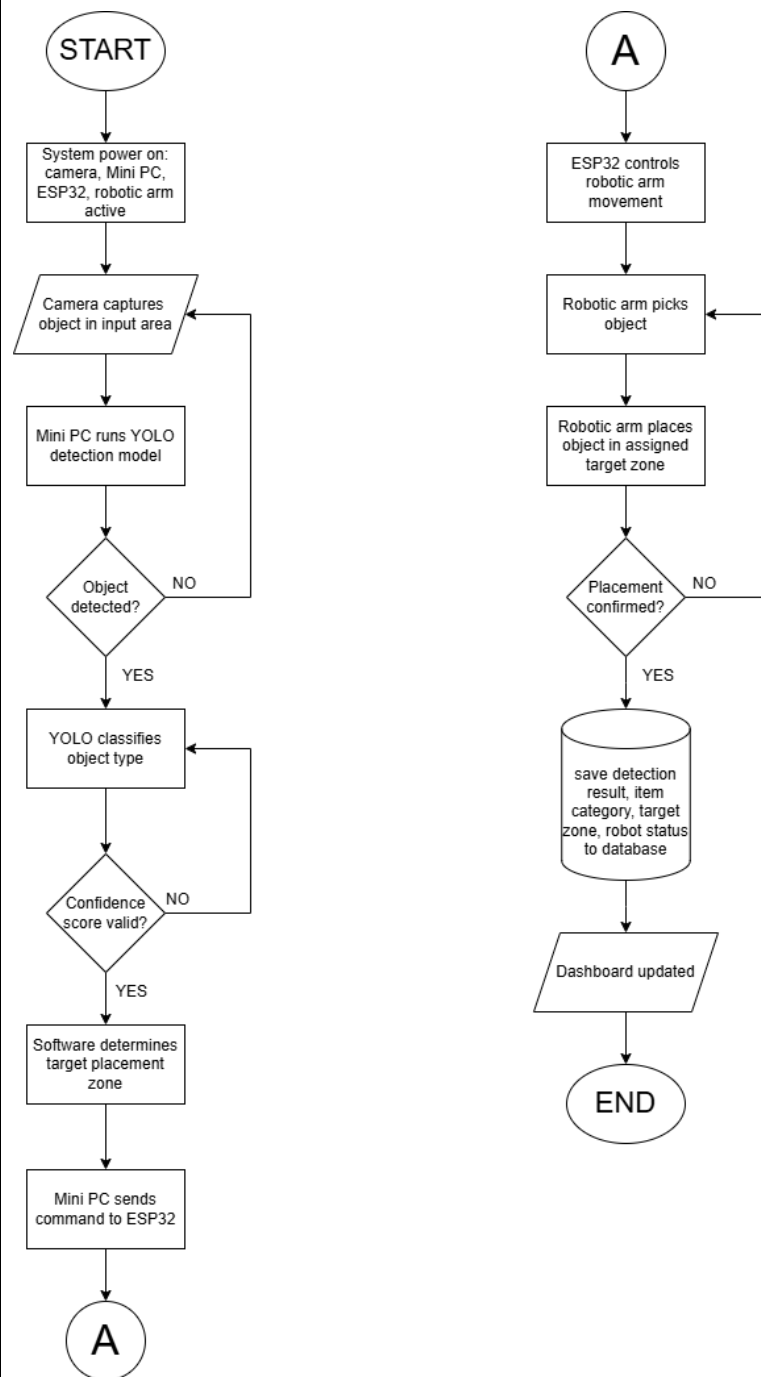


Wiring Diagram Description

The wiring architecture of the Smart Safety Locker System consists of five main pathways: power distribution, input, communication, output control, and cloud/IoT connectivity. These pathways integrate the hardware components with the software system to support real-time data acquisition, object detection, command transmission, actuator control, database storage, and dashboard updates. In this system, the Mini PC acts as the main software processing unit, while the ESP32 functions as the hardware controller that executes commands from the software.

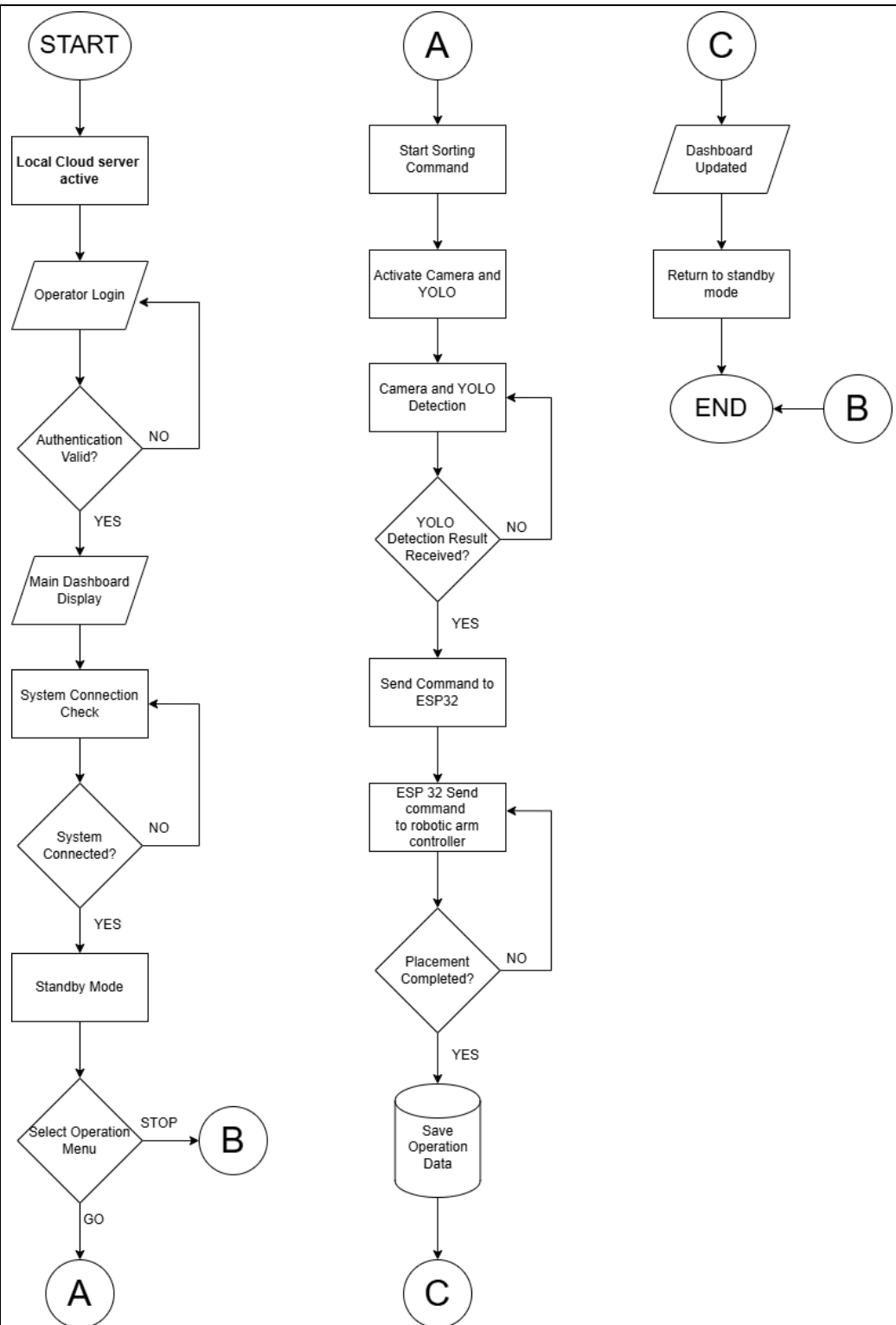
The input process uses an Advantech iCam-300 camera to capture real-time images from the locker input area. The captured image data is sent to the Mini PC, where computer vision software processes the data for object detection, item classification, and verification. After the software validates the result, the Mini PC sends commands to the ESP32 through a bidirectional communication pathway. The ESP32 also sends actuator feedback and system status back to the Mini PC to ensure synchronization between software decision-making and hardware execution.

The ESP32 controls the output devices, including stepper motors and servo motors, based on commands received from the Mini PC. The cloud and IoT connectivity pathway allows users to access the system through smartphones or laptops, while the cloud server manages user data, locker requests, transaction records, and system status. The dashboard is updated in real time, enabling remote monitoring, centralized control, and easier system maintenance through a cloud-based platform.



Flowchart 1. Software-Based Workflow for Robotic Arm Object Sorting System

This flowchart represents the software architecture workflow of the proposed automated object detection and sorting system. The system uses a camera as the input sensor, a Mini PC as the main processing unit, a YOLO model for object detection and classification, an ESP32 as the hardware controller, and a robotic arm as the actuator. The software controls the entire process, starting from image acquisition, object detection, confidence score validation, target zone determination, command transmission, placement verification, database storage, and dashboard update. Through this workflow, the system is able to operate automatically while still providing error handling mechanisms such as repeated scanning, warning notifications, operator confirmation, retry actions, and placement validation.



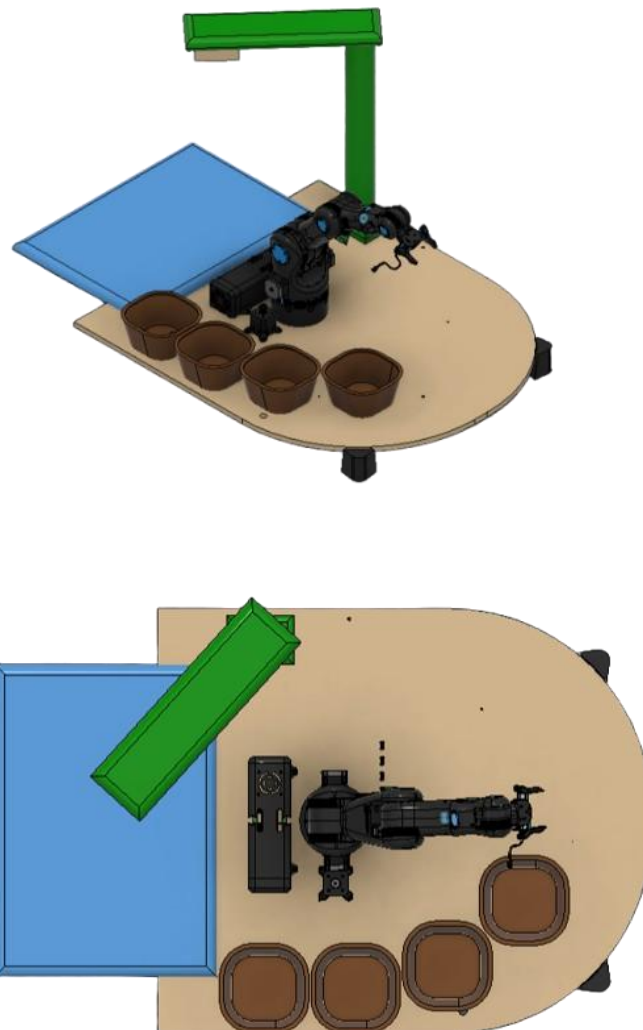
Flowchart 2. Software Workflow of Web Dashboard and Robotic Arm Control System

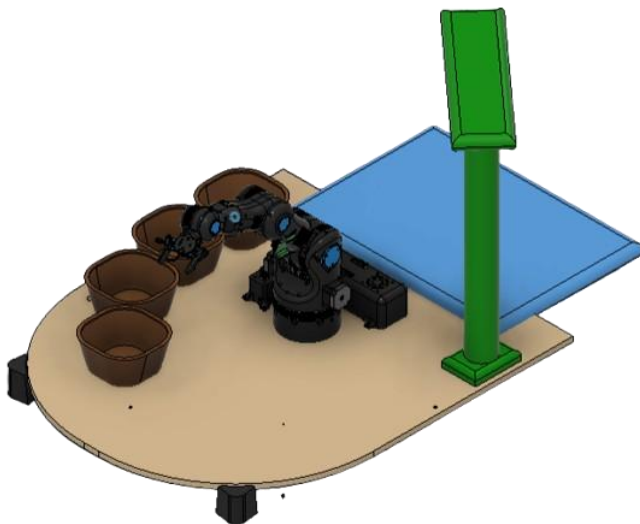
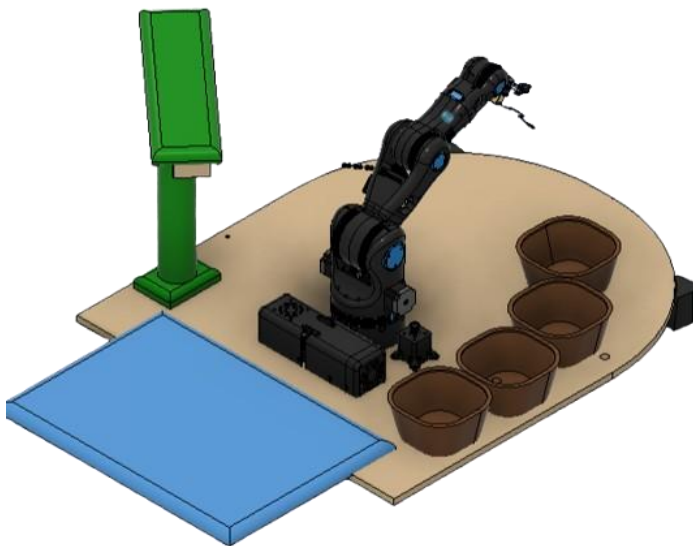
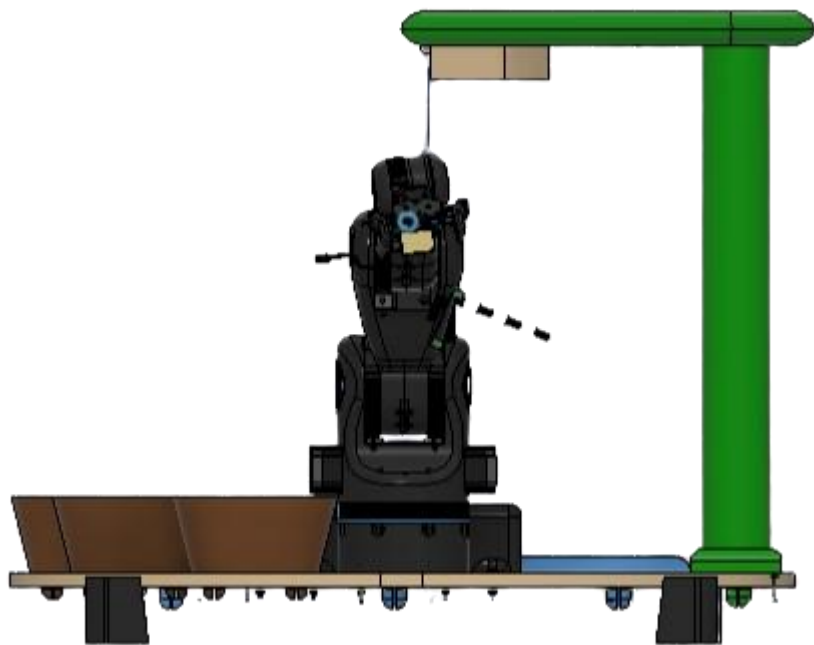
The flowchart illustrates the application and dashboard workflow of the YOLO-based robotic arm control system. The process starts with server activation, followed by operator login and authentication. After successful authentication, the dashboard displays real-time information such as camera feed, detected object class, confidence score, target placement zone, robotic arm status, ESP32 connection status, operation history, and error notifications.

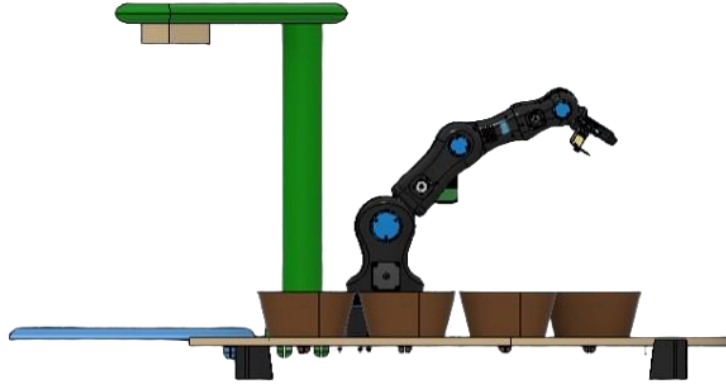
The system then checks the connection between the dashboard, Mini PC, camera, YOLO detection service, database, and ESP32 controller. When the connection is stable, the operator can access the operation menu and start the Auto Sorting Mode. In this mode, the camera and YOLO model detect and classify the object, while the software validates the confidence score and determines the target placement zone.

Once the detection result is valid, the application sends a command to the ESP32, which controls the robotic arm movement. The dashboard displays the robotic arm operation status in real time and provides retry or manual override options if an error occurs. After the placement is completed, the system saves the operation data into the database and updates the dashboard. This flowchart shows that the software plays a central role in managing authentication, object detection, decision-making, device communication, data storage, and real-time monitoring.

3D Prototype Design







1. Robotic Arm Design

At the center of the platform is a 5-DOF robotic arm with a matte black finish and blue servo joint indicators. The arm consists of five interconnected sections: the base, lower arm, upper arm, wrist, and gripper end-effector. These joints enable flexible movement across the operational workspace. The robotic arm is mounted on a compact motor housing fixed securely to the platform surface. During operation, the arm performs multiple movement configurations, including resting, object transfer, and item pickup or release from the designated input area.

2. Input Tray Zone Design

The input zone is positioned along the curved section of the platform and consists of several dark brown bucket-shaped trays. These trays serve as item drop-off points for users before the storage process begins. The trays are arranged within the maximum reach range of the robotic arm to ensure efficient object transfer without excessive joint movement. Their organized placement supports smooth robotic operation and improves workspace accessibility.

3. ASRS Gantry Structure Design The ASRS gantry structure is located on the straight side of the platform and is visually distinguished by its bright green frame. The structure consists of a vertical support pillar connected to a horizontal beam, forming an inverted L-shaped configuration. This gantry functions as the primary support structure for the ASRS mechanism and represents the elevated storage area of the system. A small storage box positioned on the beam demonstrates the item placement process within the automated storage subsystem.

4. Linear Slide Rail Design

Attached to the base of the ASRS gantry is a blue linear slide platform that represents the travel rail of the ASRS mechanism. The rail serves as the movement pathway for the storage system, enabling precise linear positioning toward designated locker slots. Its contrasting blue color clearly distinguishes the movement axis from the rest of the platform structure while emphasizing the directional motion of the ASRS subsystem.

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Enabling an Intelligent Planet

Advantech AIoT InnoWorks - Project Information: Team 2			
Team Name		No. of Team Members	
Project Topic		Adviser	
Member background (department)			
Project introduction For example: The target audience of the project, the issue want to be resolved and the expected benefits of the project			
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 <i>Enabling an Intelligent Planet</i>			